Supporting Information

Gated Bottleneck Convolution with High-Similarity-Pass Attention Enables Accurate Drosha and Dicer Cleavage Site Prediction

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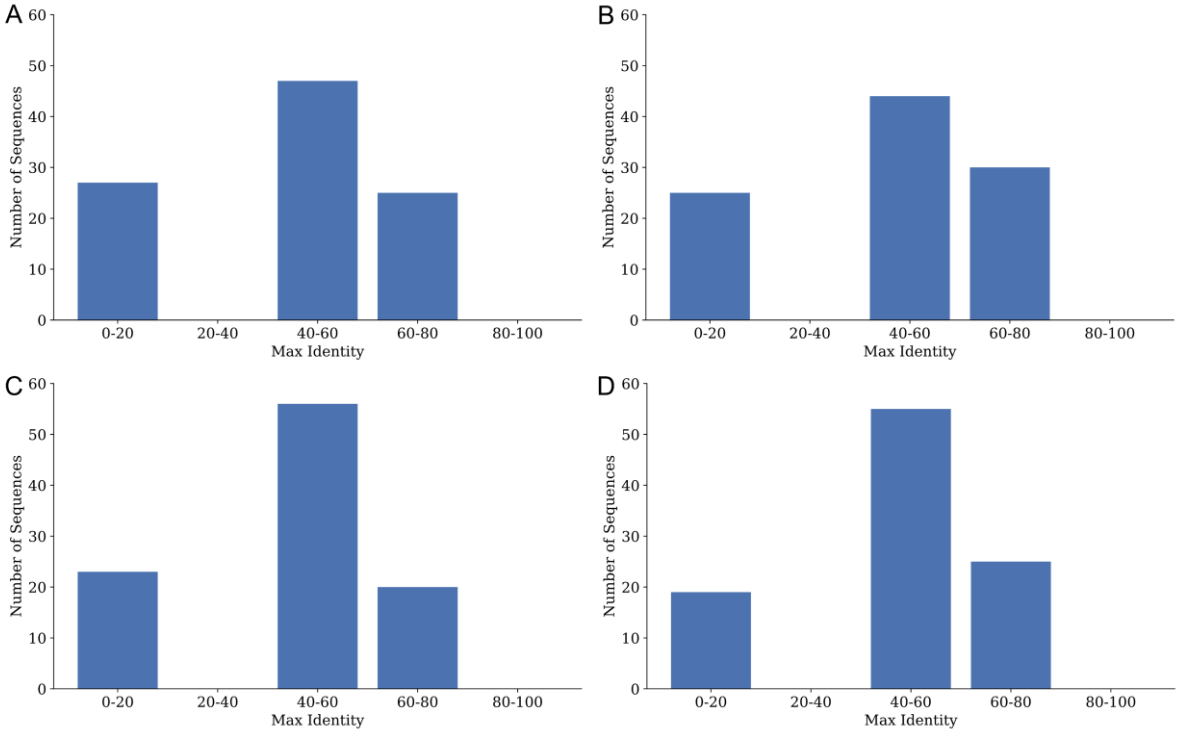
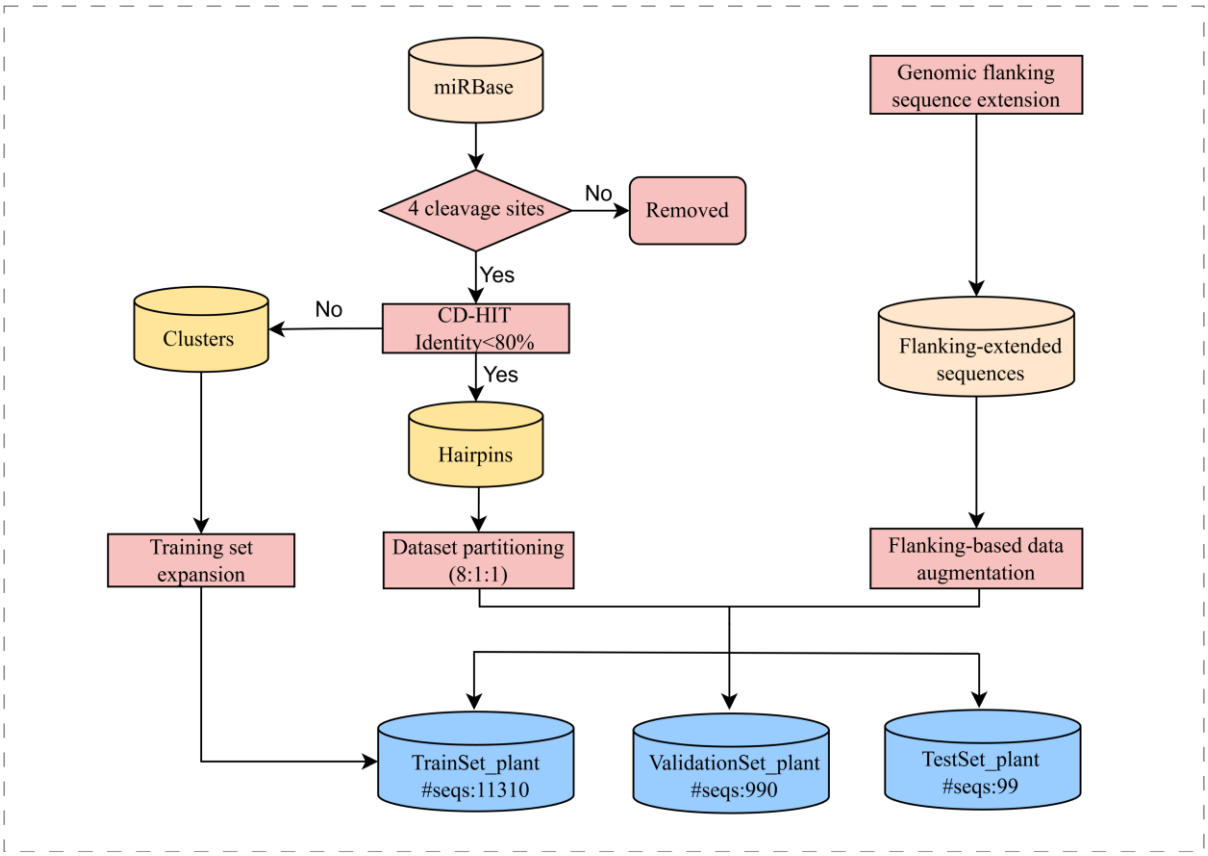


Supplementary Fig. S1. Sequence-identity comparisons among splits in the plant dataset. The test and validation sets are compared to the original training set and the training set augmented with similar miRs in terms of sequence identity. (a) Test set compared to training set. (b) Test set compared to training set with similar miRs. (c) Validation set compared to training set. (d) Validation set compared to training set with similar miRs.



Supplementary Fig. S2. Plant dataset construction pipeline.

Supplementary Table S1. Scale of animal and plant datasets.

Data split	Raw data	Enhanced data
TrainSet	8491	84910
ValidationSet	490	4900
TestSet	491	\
TrainSet*	1131	11310
ValidationSet*	99	990
TestSet*	99	\

Note: Data split denotes the partition of samples into training, validation, and test sets. Raw data indicates the number of original sequences in each data set. Enhanced data represents the number of sequences after data augmentation. Entries without an asterisk (*) are animal datasets, whereas those marked with an asterisk (*) are plant datasets.

Supplementary Table S2. Animal species genome FASTA files and sources.

Species	Genome ID	miRBase Version
Abu	GCA_000239415.1	22.1
Aca	GCA_000090745.2	22.1
Aga	GCA_000005575.1	22.1
Ame	GCA_000002195.1	22.1
Ami	GCA_000281125.4	22.1
Apl	GCA_000355885.1	22.1
Bfl	GCF_000003815.1	22.1
Bma	GCA_000002995.4	22.1
Bmo	GCA_000151625.1	22.1
Bta	GCA_000003205.6	22.1
Cbr	GCA_000004555.3	22.1
Cel	GCA_000002985.3	22.1
Cfa	GCA_000002285.2	22.1
Cgr	GCA_000223135.1	22.1
Chi	GCA_000317765.2	22.1
Cin	V2.0	22.1
Cja	GCA_000004665.1	22.1
Cli	GCA_000337935.1	22.1
Cpi	GCA_000241765.2	22.1
Cpo	GCA_000151735.1	22.1
Cqu	GCA_000209185.1	22.1
Crm	WS200	22.1
Cte	GCA_000328365.1	22.1
Dan	GCA_000005115.1	22.1
Der	Dere_r1.3_FB2010_02	22.1
Dgr	Dgri_r1.3_FB2010_02	22.1
Dme	GCA_000001215.4	22.1
Dmo	Dmoj_r1.3_FB2010_02	22.1
Dno	Dasypus novemcinctus	22.1
Dpe	Dper_r1.3_FB2010_02	22.1
Dps	Dpse_r2.26_FB2012_01	22.1
Dpu	GCA_000187875.1	22.1

Dre	GCA_000002035.4	22.1
Dse	Dsec_r1.3_FB2010_02	22.1
Dsi	GCA_000259055.1	22.1
Dvi	GCA_000005245.1	22.1
Dwi	Dwil_r1.3_FB2010_02	22.1
Dya	GCA_000005975.1	22.1
Eca	GCF_000002305.2	22.1
Eel	GCA_003665695.1	22.1
Efu	GCA_000308155.1	22.1
Egr	GCA_000524195.1	22.1
Emu	WTSI3	22.1
Fru	GCA_000180615.2	22.1
Gga	GCA_000002315.3	22.1
Ggo	GCA_000151905.3	22.1
Gmo	GCA_000231765.1	22.1
Hme	GCA_000313835.2	22.1
Hpo	NHp_1_0	22.1
Hsa	Heligmosomoides polygyrus	22.1
Lgi	GCA_000327385.1	22.1
Mco	GCA_000951315.1	22.1
Mdo	GCF_000002295.2	22.1
Mml	GCA_000772875.3	22.1
Mmu	GCA_000001635.8	22.1
Mze	GCA_000238955.3	22.1
Nbr	GCA_000239395.1	22.1
Nve	GCA_000209225.1	22.1
Oan	OANA5	22.1
Oar	GCA_000298735.2	22.1
Ocu	GCA_000003625.1	22.1
Oha	GCA_000516915.1	22.1
Ola	MEDAKA1	22.1
Oni	GCA_000188235.2	22.1
Pal	GCA_000325575.1	22.1
Pbv	GCA_000186305.2	22.1
Pma	GCA_000148955.1	22.1

Pny	GCA_000239375.1	22.1
Ppa	GCA_000258655.2	22.1
Ppc	GCA_000180635.3	22.1
Ppy	GCF_000001545.4	22.1
Prd	GCA_000341325.1	22.1
Pte	GCA_000365465.2	22.1
Ptr	GCA_000001515.5	22.1
Pxy	V1.0	22.1
Rno	GCA_000001895.4	22.1
Sfr	GCA_000753635.2	22.1
Sha	GCA_000189315	22.1
Sko	GCA_000003605.1	22.1
Sma	WTSI5	22.1
Spu	GCA_000002235.3	22.1
Ssc	GCA_000003025.4	22.1
Str	WTSIv4	22.1
Tca	Tcas4.0	22.1
Tch	GCA_000334495.1	22.1
Tgu	TaeGlu3.2.4	22.1
Tni	Tetraodon8	22.1
Xla	JGIv8.0	22.1
Xtr	GCA_000004195.3	22.1

Note. Species denotes the abbreviated species name used throughout this study. Genome ID refers to the reference genome assembly accession corresponding to each species, as provided by the genome repository from which the FASTA files were obtained. miRBase Version indicates the release of the miRBase database used for miRNA annotation; all miRNA annotations in this study were derived from miRBase version 22.1.

Supplementary Table S3. Plant species genome FASTA files and sources.

Species	Genome ID	miRBase Version
Tly	GCA_000004255.1	22.1
Ath	GCA_000001735.1	22.1
Bdi	GCA_000005505.1	22.1
Bra	GCA_000309985.1	22.1
Cre	GCA_000002595.2	22.1
Csa	CSAV2.0	22.1
Csi	GCA_000317415.1	22.1
Fve	V1.1	22.1
Gma	GCA_000004515.3	22.1
Gra	GCA_000327365.1	22.1
Lja	Lj2.5	22.1
Mes	Cassava4	22.1
Mtr	JCVI_Mt3.5.2	22.1
Osa	MSU7	22.1
Ppe	GCA_000346465.2	22.1
Ptc	GCA_000002775.2	22.1
Pvu	GCA_000499845.1	22.1
Sbi	GCA_000003195.3	22.1
Sly	GCA_000188115.2	22.1
Stu	GCA_000226075.1	22.1
Vvi	GCA_000003745.2	22.1
Zma	GCA_000005005.6	22.1

Note. Column definitions are the same as those in **Table S2**.

Supplementary Table S4. Composition of the Integrated Human Pre-miRNA Case Panel (IHPCP).

Pre-miRNA ID	Disease relevance	Evidence source
hsa-mir-15a	Chronic lymphocytic leukaemia	PMID: 24732594
hsa-mir-27a	Bipolar disorder	PMID: 35379867
hsa-mir-30c-1	Gastric cancer	PMID: 22108846
hsa-mir-101-2	Breast cancer	PMID: 24475105
hsa-mir-124-1	Spinal cord injury	PMID: 36160329
hsa-mir-125a	Invasive breast cancer	PMID: 36516216
hsa-mir-126	Platelet activation	PMID: 26646931
hsa-mir-142	Cancer	PMID: 33916692
hsa-mir-145	Cancer	PMID: 31767308
hsa-mir-146a	Cancer, inflammation, cardiovascular disease	PMID: 32994184; PMID: 18474871
hsa-mir-149	Coronary artery disease	PMID: 32092393
hsa-mir-196a-2	Lung cancer, liver cancer, etc., type 2 diabetes	PMID: 33925232; PMID: 29535537
hsa-mir-423	Breast cancer, colorectal cancer	PMID: 31101240
hsa-mir-499a	Myocardial infarction, heart failure	PMID: 30102014
hsa-mir-605	Breast cancer	PMID: 30135399

Note: Pre-miRNA ID follows miRBase nomenclature.